SOME MAXIMAL INEQUALITIES.

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1. In this note, we extend the Hardy-Littlewood maximal theorem to the case of l^p-valued functions. Our results unify some classical inequalities of Hardy-Littlewood and Marcinkiewicz, as well as a variant of the maximal theorem, which Carleson used in [3]. Our version of the maximal theorem is the following.

THEOREM 1. *Let $f=(f_1, f_2, \cdots)$ be a sequence of functions on $\boldsymbol{R}^n$. Form the sequence f^*, whose k-th term is the maximal function of f_k. Then*

$$(1)\quad \|(\sum_{k=1}^{\infty}|f_k^*(\cdot)|^r)^{1/r}\|_p \leqq A_{r,p}\|(\sum_{k=1}^{\infty}|f_k(\cdot)|^r)^{1/r}\|_p \qquad (1<r,p<\infty).$$

$$(2)\quad |\{x\in R^n: (\sum_{k=1}^{\infty}|f_k^*(x)|^r)^{1/r}>\alpha\}| \leqq \frac{A_r}{\alpha}\|(\sum_{k=1}^{\infty}|f_k(\cdot)|^r)^{1/r}\|_1 \qquad (1<r<\infty).$$

$$(3)\quad \textit{If } (\sum_{k=1}^{\infty}|f_k(\cdot)|^r)^{1/r}\in L^\infty(\boldsymbol{R}^n), \textit{ then } (\sum_{k=1}^{\infty}|f_k^*(\cdot)|^r)^{1/r}\in \exp(L^r) \qquad (1<r<\infty).$$

The maximal function f_k^* is defined by $f_k^*(x)=\sup\frac{1}{|Q|}\int_Q |f_k(y)|\,dy$, where the "sup" is taken over all cubes Q centered at x.

If we put $f_1=f$, $f_2=f_3=\cdots 0$, then of course we recover the usual maximal theorem (see Wiener [7]). On the other hand, suppose we put $f_k=\chi_{Q_k}$, the characteristic function of Q_k, where $Q_1, Q_2, \cdots$ are disjoint cubes in $\boldsymbol{R}^n$. Then $(\sum_{k=1}^{\infty}|f_k^*(x)|^r)\approx\sum_k \frac{|Q_k|^r}{(|x+y_k|^n+|Q_k|)^r}\equiv \mathcal{J}_r(x)$, where y_k is the center of Q_k. This expression is a simple modification of the classical Marcinkiewicz integral of order r, corresponding to the cubes $\{Q_k\}$ (see Calderón and Zygmund [1] and also Carleson [4]). Our maximal theorem asserts that

$$(1)\quad \text{For } \frac{1}{r}<q<\infty, \int_{\boldsymbol{R}^n}(\mathcal{J}_r(x))^q\,dx \leqq A_{q,r}\cdot\sum_k |Q_k|.$$

Received March 16, 1970.

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(2) $|\{x \in R^n : \mathcal{J}_r(x) > \alpha\}| \leqq \frac{A_r}{\alpha^{1/r}} \cdot \sum_k |Q_k|$.

(3) $\mathcal{J}_r(\cdot)$ belongs to the exponential class.

Inequalities (1) and (3) are the usual estimates for the Marcinkiewicz integral—(2) seems to be new. The point is that our maximal theorem unifies the classical inequalities for the maximal function, and those for the Marcinkiewicz integral.

2. Proof of Theorem 1. We look separately at the cases $p = r$, $p < r$, and $p > r$. For the case $p = r$, inequality (1) follows immediately from the usual maximal theorem, since

$$\int_{R^n} (\sum_{k=1}^{\infty} |f_k^*(x)|^r)\, dx = \sum_{k=1}^{\infty} \int_{R^n} |f_k^*(x)|^r\, dx \leqq A_r \sum_{k=1}^{\infty} \int_{R^n} |f_k(x)|^r\, dx$$

$$= A_r \int_{R^n} (\sum_{k=1}^{\infty} |f_k(x)|^r)\, dx.$$

Using the known case $p = r$, we can now prove inequality (2), as follows. Let $f = (f_1, f_2, \cdots)$ and $\alpha > 0$ be given. Apply the Calderón-Zygmund lemma (see [2], p. 91) to $(\sum_{k=1}^{\infty} |f_k(\cdot)|^r)^{1/r}$ to obtain a collection $\{Q_j\}$ of pairwise disjoint cubes in R^n, with the properties

(a) $\sum_j |Q_j| \leqq \frac{A}{\alpha} \| (\sum_{k=1}^{\infty} |f_k(\cdot)|^r)^{1/r} \|_1$.

(b) $(\sum_{k=1}^{\infty} |f_k(x)|^r)^{1/r} \leqq \alpha$ if $x \notin \Omega \equiv \bigcup_j Q_j$.

(c) $\frac{1}{|Q_j|} \int_{Q_j} (\sum_{k=1}^{\infty} |f_k(y)|^r)^{1/r}\, dy \leqq A\alpha$ for each Q_j.

Then we can decompose f_k as $f_k = f_k' + f_k''$, where $f_k' = f_k \cdot \chi_{R^n - \Omega}$ and $f_k'' = f_k \cdot \chi_{\Omega}$. Since

$$(\sum_{k=1}^{\infty} |f_k^*(\cdot)|^r)^{1/r} \leqq (\sum_{k=1}^{\infty} |f_k'^*(x)|^r)^{1/r} + (\sum_{k=1}^{\infty} |f_k''^*(x)|^r)^{1/r},$$

inequality (2) will be proved if we can show that

(4) $|\{x \in \boldsymbol{R}^n : (\sum_{k=1}^{\infty} |f_k'^*(x)|^r)^{1/r} > \alpha\}| \leqq \frac{A}{\alpha} \| (\sum_{k=1}^{\infty} |f_k(\cdot)|^r)^{1/r} \|_1$

and

(5) $|\{x \in \boldsymbol{R}^n : (\sum_{k=1}^{\infty} |f_k''^*(x)|^r)^{1/r} > \alpha\}| \leqq \frac{A}{\alpha} \| (\sum_{k=1}^{\infty} |f_k(\cdot)|^r)^{1/r} \|_1$.

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First, let us prove (4). From inequality (b) and the obvious fact that

$$\| (\sum_{k=1}^{\infty} | f_k'(\cdot)|^r)^{1/r} \|_1 \leqq \| (\sum_{k=1}^{\infty} | f_k(\cdot)|^r)^{1/r} \|_1$$

we see

$$\| (\sum_{k=1}^{\infty} | f_k'(\cdot)|^r)^{1/r} \|_r^r \leqq \alpha^{r-1} \| (\sum_{k=1}^{\infty} | f_k(\cdot)|^r)^{1/r} \|_1.$$

Therefore, by the easy case $p = r$ of inequality (1),

$$\| (\sum_{k=1}^{\infty} | f_k'^*(\cdot)|^r)^{1/r} \|_r^r \leqq A_r \alpha^{r-1} \| (\sum_{k=1}^{\infty} | f_k(\cdot)|^r)^{1/r} \|_1.$$

This yields inequality (4) immediately, by the Chebyshev inequality.

To prove (5), we define functions $\tilde{f}_k$ by setting

$$\tilde{f}_k(x) = \begin{cases} \frac{1}{|Q_j|} \int_{Q_j} |f_k(y)|\, dy \text{ if } x \in Q_j \\ 0 \text{ if } x \notin \Omega. \end{cases}$$

For $x \in Q_j$,

$$(\sum_{k=1}^{\infty} | \tilde{f}_k(x)|^r)^{1/r} = (\sum_{k=1}^{\infty} [\frac{1}{|Q_j|} \int_{Q_j} |f_k(y)|\, dy]^r)^{1/r}$$

$$\leqq \frac{1}{|Q_j|} \int_{Q_j} (\sum_{k=1}^{\infty} | f_k(y)|^r)^{1/r}\, dy,$$

by the vector-valued form of Minkowski's inequality, $\leqq A\alpha$ by inequality (c). For $x \notin \Omega$, all $\tilde{f}_k$ are zero, so $(\sum_{k=1}^{\infty} | \tilde{f}_k(x)|^r)^{1/r} = 0$. Thus the function $(\sum_{k=1}^{\infty} | \tilde{f}_k(\cdot)|^r)^{1/r}$ has support in Ω and is bounded by $A\alpha$, which implies that

$$\| (\sum_{k=1}^{\infty} |\tilde{f}_k(\cdot)|^r)^{1/r} \|_r^r \leqq A\alpha^r |\, \Omega\, | \leqq A\alpha^{r-1} \| (\sum_{k=1}^{\infty} |f_k(\cdot)|^r)^{1/r} \|_1.$$

As in the proof of (4), we now have from case $p = r$ of (1) and from the Chebyshev inequality, that

$$(6) \qquad |\{x \in \boldsymbol{R}^n : (\sum_{k=1}^{\infty} |\tilde{f}_k^*(x)|^r)^{1/r} > \alpha\}| \leqq \frac{A}{\alpha} \| (\sum_{k=1}^{\infty} | f_k(\cdot)|^r)^{1/r} \|_1.$$

We shall deduce (5) from (6) by showing that for most points $x \in \boldsymbol{R}^n$, $f_k''^*(x) \leqq A\tilde{f}_k^*(x)$. This will complete the proof of our weak-type result (2).

For any cube $Q \subseteq \boldsymbol{R}^n$, let $\tilde{Q}$ denote a cube concentric with Q, but having diameter $2n$ times as large. Set $\tilde{\Omega} = \bigcup_j \tilde{Q}_j$. Obviously

$$|\, \tilde{\Omega}\, | \leqq \frac{A}{\alpha} \| (\sum_{k=1}^{\infty} | f_k(\cdot)|^r)^{1/r} \|_1, \text{ by (a).}$$

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We claim that for $x \notin \tilde{\Omega}$, $f_k''^*(x) \leqq A\tilde{f}_k^*(x)$. In fact,

$$f_k''^* \approx \sup_{Q \text{ contains } x}\{\frac{1}{|Q|}\int_Q |f_k''(y)|\,dy\},$$

and for any fixed Q containing x,

$$\frac{1}{|Q|}\int_Q |f_k''(y)|\,dy = \frac{1}{|Q|}\sum_{j\in J}\int_{Q_j\cap Q}|f_k''(y)|\,dy,$$

where $J = \{j \mid Q_j \cap Q \neq \emptyset\}$. On the other hand, $Q_j \cap Q \neq \emptyset$ and $x \in Q - \tilde{\Omega} \subseteq Q - \tilde{Q}_j$ imply that $Q_j \subseteq \tilde{Q}$, by the geometry of $\boldsymbol{R}^n$. Therefore,

$$\begin{aligned}\frac{1}{|Q|}\sum_{j\in J}\int_{Q_j\cap Q}|f_k''(y)|\,dy &\leqq \frac{1}{|Q|}\sum_{j\in J}\int_{Q_j}|f_k''(y)|\,dy \\ &= \frac{1}{|Q|}\sum_{j\in J}\int_{Q_j}|\tilde{f}_k(y)|\,dy \leqq \frac{1}{|Q|}\int_Q |\tilde{f}_k(y)|\,dy \\ &\leqq \frac{A}{|\tilde{Q}|}\int_{\tilde{Q}}|\tilde{f}_k(y)|\,dy \leqq A\tilde{f}_k^*(x).\end{aligned}$$

We conclude that for $x \in \boldsymbol{R}^n - \tilde{\Omega}$, $\frac{1}{|Q|}\int_Q |f_k''(y)|\,dy \leqq A\tilde{f}_k(x)$, (any Q containing x), which implies that $f_k''^*(x) \leqq A\tilde{f}_k^*(x)$.

Now we can deduce (5) from (6). For,

$$\begin{aligned}&|\{x \in \boldsymbol{R}^n - \tilde{\Omega} : (\sum_{k=1}^{\infty}|f_k''^*(x)|^r)^{1/r} > A\alpha\}| \\ &\qquad \leqq |\{x \in \boldsymbol{R}^n - \tilde{\Omega} : (\sum_{k=1}^{\infty}|\tilde{f}_k^*(x)|^r)^{1/r} > \alpha\}| \\ &\qquad \leqq \frac{A}{\alpha}\|(\sum_{k=1}^{\infty}|f_k(\cdot)|^r)^{1/r}\|_1.\end{aligned}$$

Since also $|\tilde{\Omega}| \leqq \frac{A}{\alpha}\|(\sum_{k=1}^{\infty}|f_k(\cdot)|^r)^{1/r}\|_1$, we have inequality (5). Since we had reduced the whole proof of (2) to the inequality (5), (2) follows.

For $p \leqq r$, inequality (1) is a simple consequence of the case $p = r$ and of inequality (2), by the Marcinkiewicz interpolation theorem. Thus, to prove Theorem 1, we have only to demonstrate (3) and case $p \geqq r$ in (1).

3. Weighted maximal functions. To prove the remaining parts of our maximal theorem, we are going to use duality. That is, we shall control the size of $(\sum_{k=1}^{\infty}|f_k^*(\cdot)|^r)^{1/r}$ by estimating the integral

$$\int_{\boldsymbol{R}^n}(\sum_{k=1}^{\infty}|f_k^*(x)|^r)\phi(x)\,dx$$

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for all ϕ belonging to a suitable space of test functions. To estimate such an integral, we use the following inequality.

LEMMA 1. *Let f and ϕ be positive real-valued functions on $\mathbf{R}^n$. Then for $r > 1$, $\int_{\mathbf{R}^n} (f^*(x))^r \phi(x)\,dx \leqq B_r \int_{\mathbf{R}^n} |f(x)|^r \phi^*(x)\,dx$, with B_r depending only on r.*

We prove the lemma first for a closely related maximal function f^{**} instead of f^*. We define f^{**} as follows: Divide $\mathbf{R}^n$ into a grid of cubes, each of length 2^k, so that the vertices of each cube have coordinates which are integral multiples of 2^k: k is an integer, positive or negative. Call the resulting family of cubes $\mathcal{F}_k$. Notice that the cubes in $\mathcal{F}_k$ arise from those in $\mathcal{F}_{k+1}$ by bisecting the edges. Define $f^{**}(x) = \sup_{-\infty < k < \infty} \{\frac{1}{|Q|}\int_Q |f(y)|\,dy$, for all $Q \in \mathcal{F}_k$ with $x \in Q\}$.

We shall begin by proving

$$\int_{\mathbf{R}^n} (f^{**}(x))^r \phi(x)\,dx \leqq B_r \int_{\mathbf{R}^n} |f(x)|^r \phi^*(x)\,dx. \tag{7}$$

We do this as follows: Hold ϕ fixed, and look at the mapping $T: f \to f^{**}$. Then (7) says that T is bounded from $L^r(\mathbf{R}^n, \phi^*(x)\,dx)$ to $L^r(\mathbf{R}^n, \phi(x)\,dx)$. Clearly, T is bounded from $L^\infty(\mathbf{R}^n, \phi^*(x)\,dx$ to $L^\infty(\mathbf{R}^n, \phi(x)\,dx)$. If we can show that T has weak-type $(1,1)$, then our Lemma will follow immediately, by the Marcinkiewicz interpolation theorem. A weak-type inequality for T says that

$$\int_{\{x \in \mathbf{R}^n | f^{**}(x) > \alpha\}} \phi(y)\,dy \leqq \frac{A}{\alpha} \int_{\mathbf{R}^n} f(x)\phi^*(x)\,dx.$$

This is not hard to prove. For, the proof of the Calderón-Zygmund lemma of [2] shows that $\{x \in R^n \mid f^{**}(x) > \alpha\} = \bigcup_j Q_j$, where the Q_j are pairwise disjoint cubes satisfying the condition $\frac{1}{|Q_j|}\int_{Q_j} f(x)\,dx \approx \alpha$. Restricting attention to Q_j for a moment, we see that

$$\int_{Q_j} f(x)\phi^*(x)\,dx \geqq \int_{Q_j} f(x) \left[\frac{1}{|Q_j|}\int_{Q_j} \phi(y)\,dy\right] dx = \left[\int_{Q_j} \phi(y)\,dy\right] \cdot \frac{1}{|Q_j|}\int_{Q_j} f(x)\,dx \approx \alpha \cdot \int_{Q_j} \phi(y)\,dy.$$

Summing over j, we obtain

$$\alpha \cdot \int_{\{x \in \mathbf{R}^n | f^{**}(x) > \alpha\}} \phi(y)\,dy \leqq A \cdot \int_{\{x \in \mathbf{R}^n | f^{**}(x) > \alpha\}} f(x)\phi^*(x)\,dx \leqq A \int_{\mathbf{R}^n} f(x)\phi^*(x)\,dx,$$

which is exactly what we wanted. This proves (7).

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We next define $f^*(x,N) = \sup \frac{1}{|Q|}\int_Q |f(y)|\,dy$, where the sup is taken over all cubes centered at x, but whose edges have length less than N. Observe that $f^*(x,N) \uparrow f^*(x)$, *as* $N \to \infty$. We shall show that

$$(7') \qquad \int_{\boldsymbol{R}^n} (f^*(x,N))^r \phi(x)\,dx \leqq B_r' \int_{\boldsymbol{R}^n} |f(x)|^r \phi^*(x)\,dx,$$

with B_r' independent of N. To prove (7') it clearly suffices by homogeneity, to prove it for $N = 1$. We derive this case from (7) by making the elementory observation

$$f^*(x,1) \leqq c \int_{Q_0} f_t^{**}(x)\,dt$$

where Q_0 is the unit cube in $\boldsymbol{R}^n$, and $f_t^{**}(x) = (\tau_t f)^{**}(x+t)$, with $(\tau_t f)(x) = f(x-t)$. Once this observation is made (7') follows from (7), and thus the lemma is completely proved.

Now we can return to our duality argument. For any positive functions ϕ and $f_1, f_2, \cdots$, we have

$$(8) \qquad \begin{aligned} \int_{\boldsymbol{R}^n} \Big(\sum_{k=1}^{\infty} |f_k^*(x)|^r\Big)\phi(x)\,dx &= \sum_{k=1}^{\infty} \int_{\boldsymbol{R}^n} (f_k^*(x))^r \phi(x)\,dx \\ &\leqq B_r \cdot \sum_{k=1}^{\infty} \int_{\boldsymbol{R}^n} (f_k(x))^r \phi^*(x)\,dx = B_r \int_{\boldsymbol{R}^n} \Big(\sum_{k=1}^{\infty} |f_k(x)|^r\Big)\phi^*(x)\,dx. \end{aligned}$$

If in (8), we let ϕ run over the unit ball of $L^q(R^n)$ $(1 < q \leqq +\infty)$ we obtain $\big\|\big(\sum_{k=1}^{\infty} |f_k^*(\cdot)|^r\big)\big\|_{q'} \leqq A_{r,q} \cdot \big\|\big(\sum_{k=1}^{\infty} |f_k(\cdot)|^r\big)\big\|_{q'}$, where $\frac{1}{q} + \frac{1}{q'} = 1$. In other words, $\big\|\big(\sum_{k=1}^{\infty} |f_k^*(\cdot)|^r\big)^{1/r}\big\|_p \leqq A_{r,p} \big\|\big(\sum_{k=1}^{\infty} |f_k(\cdot)|^r\big)^{1/r}\big\|_p$ for $r \leqq p < +\infty$, which is the case $p > r$ in (1).

By now we have proved inequalities (1) and (2), and all that remains is (3). However, if in (8) we let ϕ range over the unit ball of $L(\log^+ L)(R^n)$, we obtain

$$\begin{aligned} \big\|\big(\sum_{k=1}^{\infty} |f_k^*(\cdot)|^r\big)\big\|_{\exp L} &\leqq A_r \cdot \big\|\big(\sum_{k=1}^{\infty} |f_k(\cdot)|^r\big)\big\|_{\infty} \\ &\cdot \sup\{\|\phi^*\|_1 \mid \|\phi\|_{L\log^+ L(R^n)} \leqq 1\} \leqq A_r \big\|\big(\sum_{k=1}^{\infty} |f_k(\cdot)|^r\big\|_{\infty}, \end{aligned}$$

which is a restatement of (3). Q. E. D.

4. Remarks. By taking into account the various constants in the proof of (1) and (2), one easily checks that $A_r = 0\big(\frac{1}{r-1}\big)$ as $r \to 1+$

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and that $A_{r,p} = 0(p^{1/r})$ as $p \to \infty$ and r stays fixed. The growth of the constants $A_{r,p}$ as $p \to \infty$ is intimately connected with the limiting result (3). In this connection see also [6, Chapter IV, Section 3] for an earlier but less decisive attack on this problem.

Lemma 1 merits further discussion. First of all, it shows that if ϕ is a positive function on the line, satisfying the condition $\phi^* \leqq C\phi$ pointwise, then the mapping $f \to f^*$ is bounded on $L^p(\boldsymbol{R}^n, \phi(x)\,dx)$ $(1 < p < +\infty)$ and has weak-type $(1,1)$—a kind of Hardy-Littlewood theorem with weight functions.[1] This assertion is quite sharp. For instance, one can easily show that if $f \to f^*$ has weak-type $(1,1)$, then $\phi^* \leqq C\phi$. Moreover, if the mapping $f \to f^*$ is bounded on any $L^p(\boldsymbol{R}^n, \phi(x)\,dx)$ with $p < \infty$, it follows that $|\{x \in \boldsymbol{R}^n \mid \phi^*(x) > \alpha\}| \leqq C \cdot |\{x \in \boldsymbol{R}^n \mid C \cdot \phi(x) > \alpha\}|$ for all α.

Secondly, our lemma sheds light on a refinement of the maximal theorem, due to Carleson [3]; see also Hörmander [5]. Given a function f on $\boldsymbol{R}^1$, set $f^{**}(x,t) = \sup\{\frac{1}{|I|}\int_I |f(y)|\,dy : I$ contains x and has diameter at least $t\}$. For a given measure μ on $\boldsymbol{R}_+^2 = R^1 x(0,\infty)$, when we assert that

$$\left(\int_{\boldsymbol{R}_+^2} |f^{**}(x,t)|^p\,d\mu(x,t)\right)^{1/p} \leqq C_p\left(\int_{\boldsymbol{R}^1} |f(x)|^p\,dx\right)^{1/p} \quad (1 < p \leqq +\infty)\,?$$

Carleson shows that this inequality is valid if μ satisfies the "growth condition" $\mu(T(I)) \leqq C\,|I|$ for each interval $I \subseteq \boldsymbol{R}^1$. Here $T(I)$ denotes an equilateral triangle with the interval $I \subseteq \boldsymbol{R}^1 \subseteq \boldsymbol{R}_+^2$ as its base.

Carleson's lemma includes the Hardy-Littlewood maximal theorem as the special case $\mu =$ Lebesgue measure on $\boldsymbol{R}^1$, and was first proved by an ingenious covering lemma stronger than the Vitali lemma used to prove the maximal theorem. However, it turns out that the new maximal theorem can be deduced from the old one by a very simple argument similar to the proof of Lemma 1. In fact, we have:

LEMMA 2. *For a measure μ on $\boldsymbol{R}_+^2$, define a function ϕ on $\boldsymbol{R}^1$ by $\phi(x) = \sup_{x \in I} \frac{\mu(T(I))}{|I|}$. Then for $f \in L^p(\boldsymbol{R}^1)$ $(1 < p \leqq +\infty)$*

$$\left(\int_{\boldsymbol{R}_+^2} |f^{**}(x,t)|^p\,d\mu(x,t)\right)^{1/p} \leqq C_p\left(\int_{\boldsymbol{R}^1} |f(x)|^p \phi(x)\,dx\right)^{1/p}.$$

Proof. We interpolate between the trivial result for $p = +\infty$ and the weak-type $(1,1)$ inequality

$$\mu(\{(x,t) \in \boldsymbol{R}_+^2 \mid f^{**}(x,t) > \alpha\}) \leqq \frac{A}{\alpha} \cdot \int_{\boldsymbol{R}^1} |f(x)|\,\phi(x)\,dx.$$

[1] Strictly speaking, we proved the weak-type $(1,1)$ property for the mapping $f \to f^{**}$: the passage to f^* can be carried out in the same spirit as the proof of (7').

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To prove the weak-type inequality, let $\Omega_1 = \{x \in R^1 \mid f^*(x) > \alpha\}$ and let $\Omega_2 = \{(x,t) \in \boldsymbol{R}_+^2 \mid f^*(x,t) > \alpha\}$. The set Ω_1 is a union of disjoint intervals $\{I_j\}$, for which $\int_{I_j} |f(x)|\,dx \approx \alpha\,|I_j|$.

For each point $x \in \boldsymbol{R}^1$, we have $f^{**}(y,t) \leqq f^*(x)$ for all (y,t) lying in the sector $S_x = \{(y,t) \in R_+^2 \mid |x-y| < t\}$. Therefore $f^*(y,t) \leqq \alpha$ for all $(y,t) \in \Omega_2' = \cup\{S_x \mid x \in R^1 - \Omega_1\}$. In other words, $\Omega_2 \subseteq R_+^2 - \Omega_2'$. On the other hand, it is geometrically obvious that $R_+^2 - \Omega_2' = \bigcup_j T(I_j)$. Consequently,

$$\mu(\Omega_2) \leqq \mu\Big(\bigcup_j T(I_j)\Big) = \sum_j \frac{\mu(T(I_j))}{|I_j|} \cdot \Big(\frac{1}{\alpha}\int_{I_j} |f(x)|\,dx\Big) \leqq \frac{1}{\alpha}\sum_j \int_{I_j} |f(x)|\,\phi(x)\,dx \leqq \frac{1}{\alpha}\int_{\boldsymbol{R}^1} |f(x)|\,\phi(x)\,dx,$$

which is exactly our weak-type inequality. Q. E. D.

The above considerations extend to n dimensions, without involving any essentially new difficulties. We state the result in the closely related context of Poisson integrals. For any $f \in L^p(\boldsymbol{R}^n)$, let $u(x,y) = (P_y * f)(x)$, where $P_y(x) = \dfrac{c_n y}{(|x|^2 + y^2)^{n+1/2}}$ is the Poisson kernel. Suppose we are given a measure μ on the upper half-space $\boldsymbol{R}_+^{n+1} = \{(x,y), x \in \boldsymbol{R}^n, y > 0\}$. For any cube $Q \in R^n$ we set $T(Q) = \{(x,y) \mid x \in Q,\ 0 < y < \operatorname{diam} Q\}$, as the analogue of the triangular region $T(I)$ described above.

We let $\phi(x) = \sup_{x \in Q} \dfrac{\mu(T(Q))}{|Q|}$.

THEOREM 2.

$$\Big(\int_{\boldsymbol{R}^{n+1}} |u(x,y)|^p\,d\mu(x,y)\Big)^{1/p} \leqq C_p\Big(\int_{\boldsymbol{R}^n} |f(x)|^p\,\phi(x)\,dx\Big)^{1/p},$$

for any $1 < p \leqq \infty$, as long of course as the right side of the inequality is finite.

For the case when ϕ may be taken to be constant, see Carleson [3] and Hörmander [5].

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